

Multimodal RAG Architecture & Benchmarks

Document ID: KB-2026-MRAG-01 | Target Domain: Knowledge Retrieval Systems

1. System Overview

This document serves as a test knowledge base for evaluation of **Multimodal Retrieval-Augmented Generation (RAG)** systems. Modern RAG architectures extend beyond text-only indexing to incorporate visual representations, charts, tables, and document layout semantics.

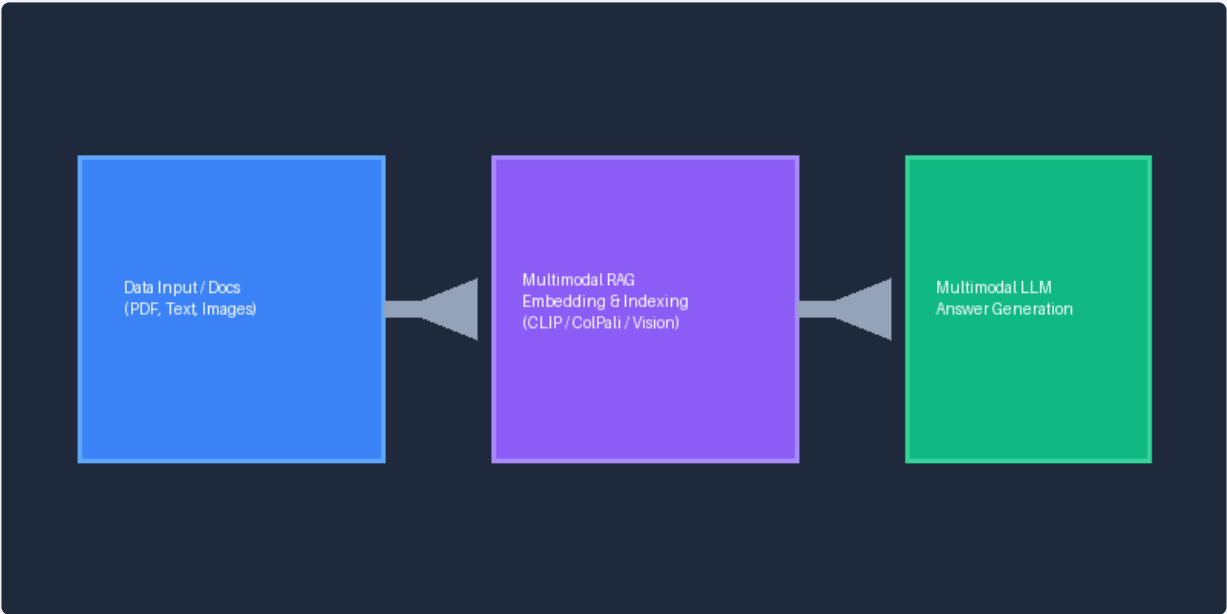


Figure 1: High-Level Pipeline of Multimodal RAG with Vision-Language Models (VLM).

Key Knowledge Fact: The system architecture shown in *Figure 1* routes data through three distinct stages: Data Input / Docs, Multimodal RAG Embedding & Indexing, and Multimodal LLM Answer Generation.

2. Visual & Textual Benchmark Data

When evaluating RAG models against visual-heavy PDFs, performance varies across text extraction, visual feature alignment, and tabular comprehension.

Component	Supported Media	Primary Vector Metric	Target Retrieval Latency
Text Processing	Plain Text, Markdown	Cosine Similarity	< 45 ms
Visual Processing	PNG, JPEG, Diagrams	Multimodal Joint Embedding Space	< 120 ms
Table Parsing	HTML Tables, Structured PDF	Hybrid Text-Visual Matching	< 85 ms

3. Mathematical Formulation

The relevance score $S(q, d)$ between query q and visual/textual document chunk d is computed using a weighted fusion model:

$$S(q, d) = \alpha \cdot \text{CosSim}(E_{\text{text}}(q), E_{\text{text}}(d)) + (1 - \alpha) \cdot \text{CosSim}(E_{\text{vision}}(q), E_{\text{vision}}(d))$$

Where $\alpha = 0.6$ represents the text modality weighting coefficient.